

CUSTOMER CHURN ANALYSIS

ABC Communications Ltd | AnalystLab Africa – Week 1

- Objective: identify customer segments and behaviours associated with churn.
- Dataset: Telco Customer Churn – 7,043 customers.
- Focus: customer profile, contract, tenure, services, payment methods and retention actions.

7.04K

Count of customerID

26.54%

Churn Rate

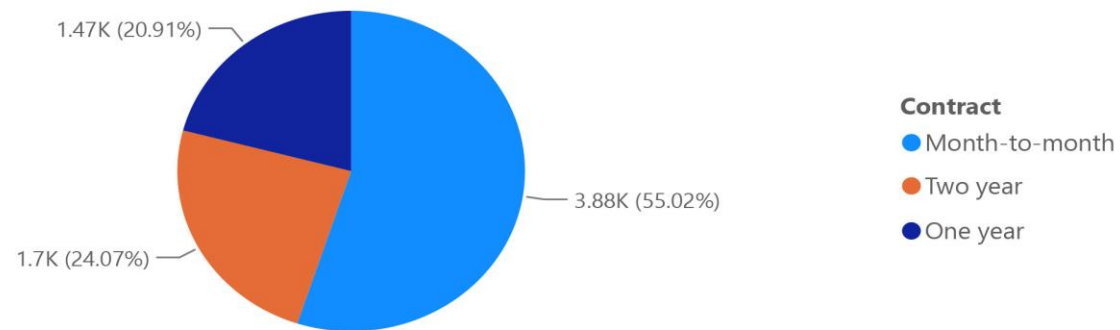
64.76

Average Monthly Charges

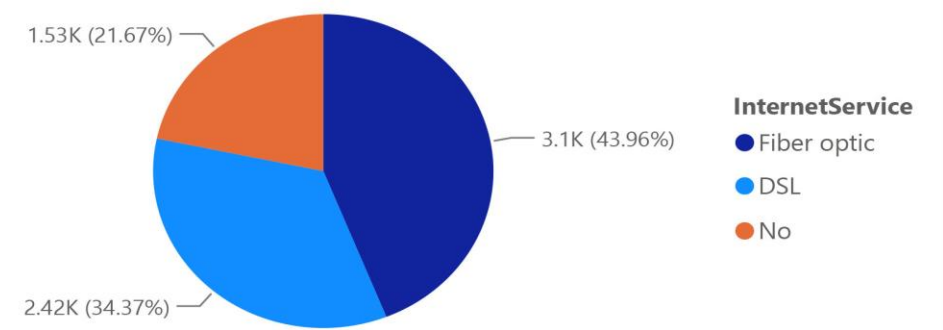
32

Average Tenure

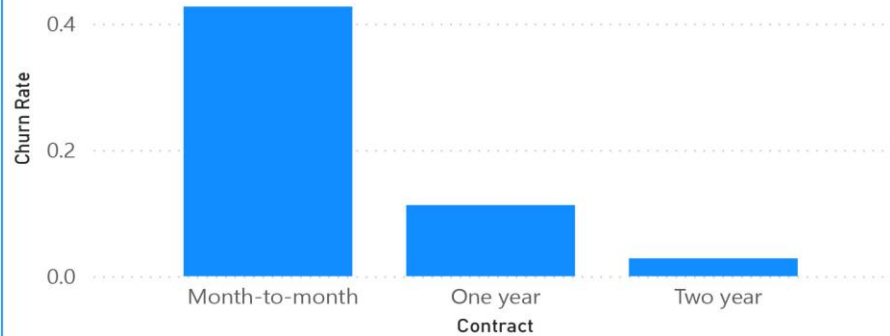
Count of customerID by Contract



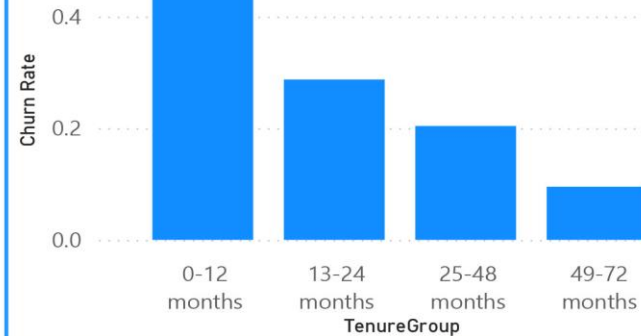
Count of customerID by InternetService



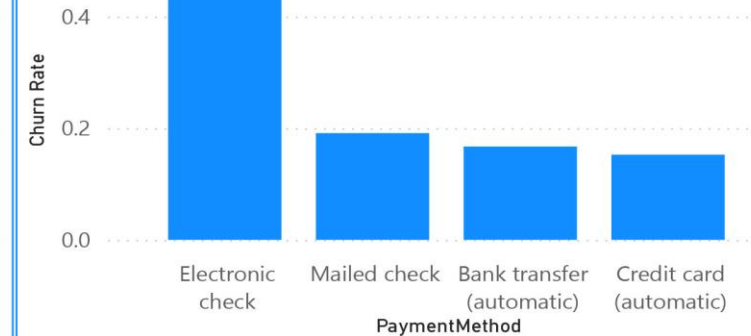
Churn Rate by Contract



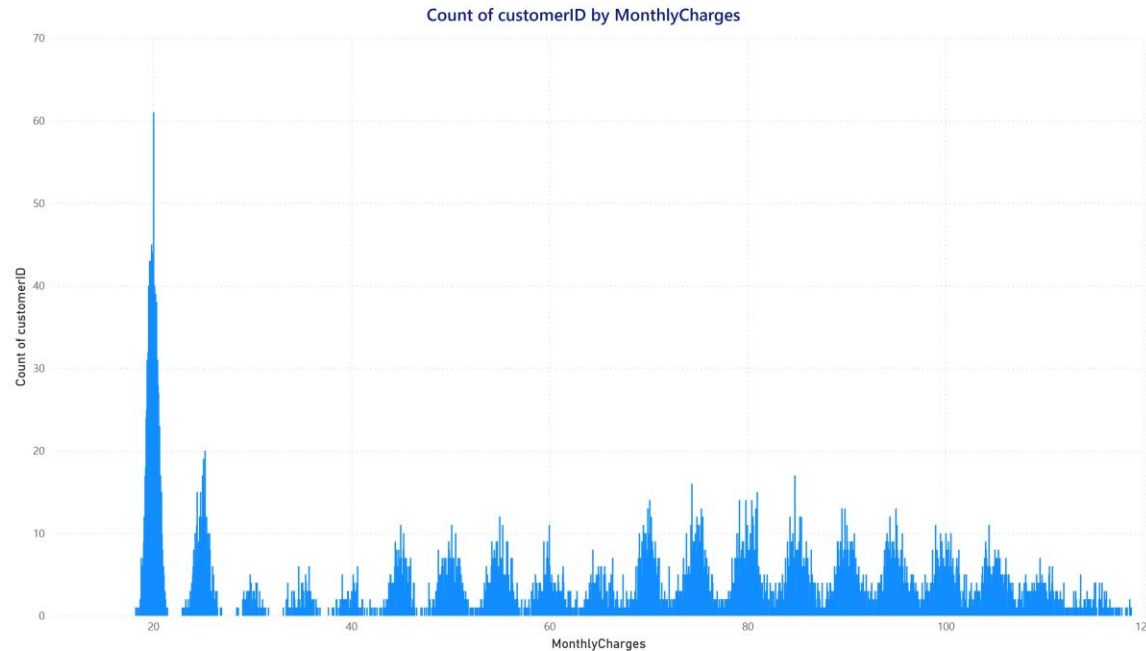
Churn Rate by TenureGroup



Churn Rate by PaymentMethod

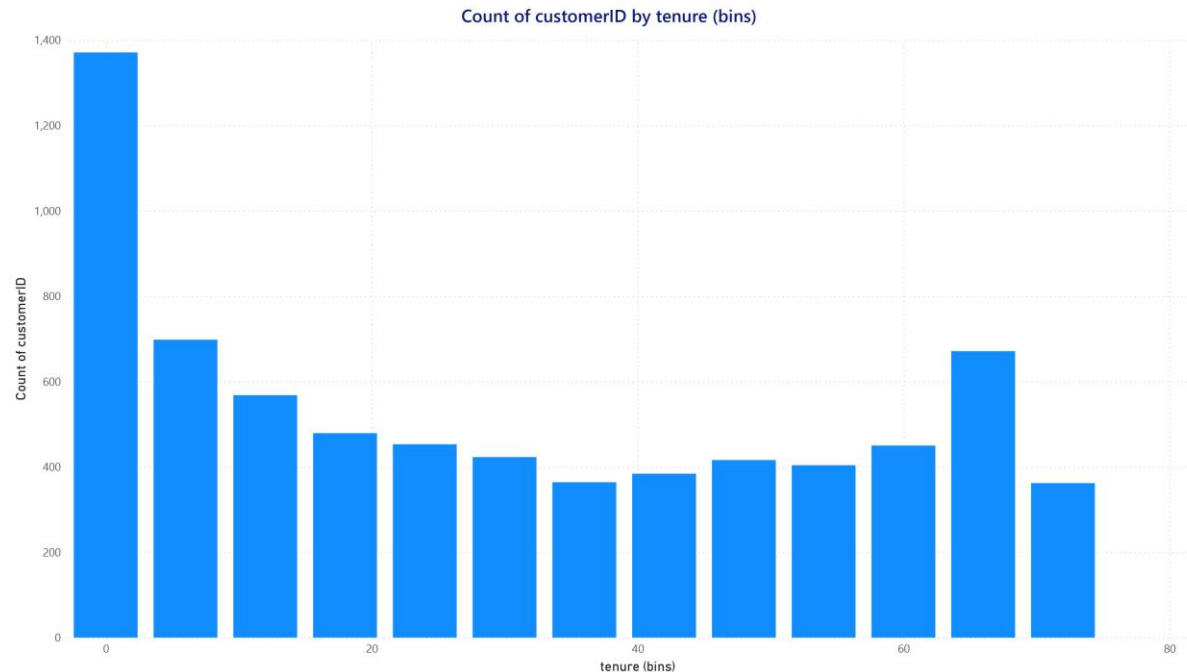


Executive Summary



- 1,869 customers churned versus 5,174 who did not; overall churn is approximately 26.5%.
- Month-to-month customers form about 55% of the base and have the highest churn rate.
- The 0–12 month tenure segment has the highest churn, highlighting early-life retention risk.
- Fiber optic customers show substantially higher churn in this dataset.
- Customers without Online Security, Online Backup, Device Protection and Tech Support show higher churn.
- Electronic check customers show substantially higher churn than other payment groups.

What Does the Customer Base Look Like?

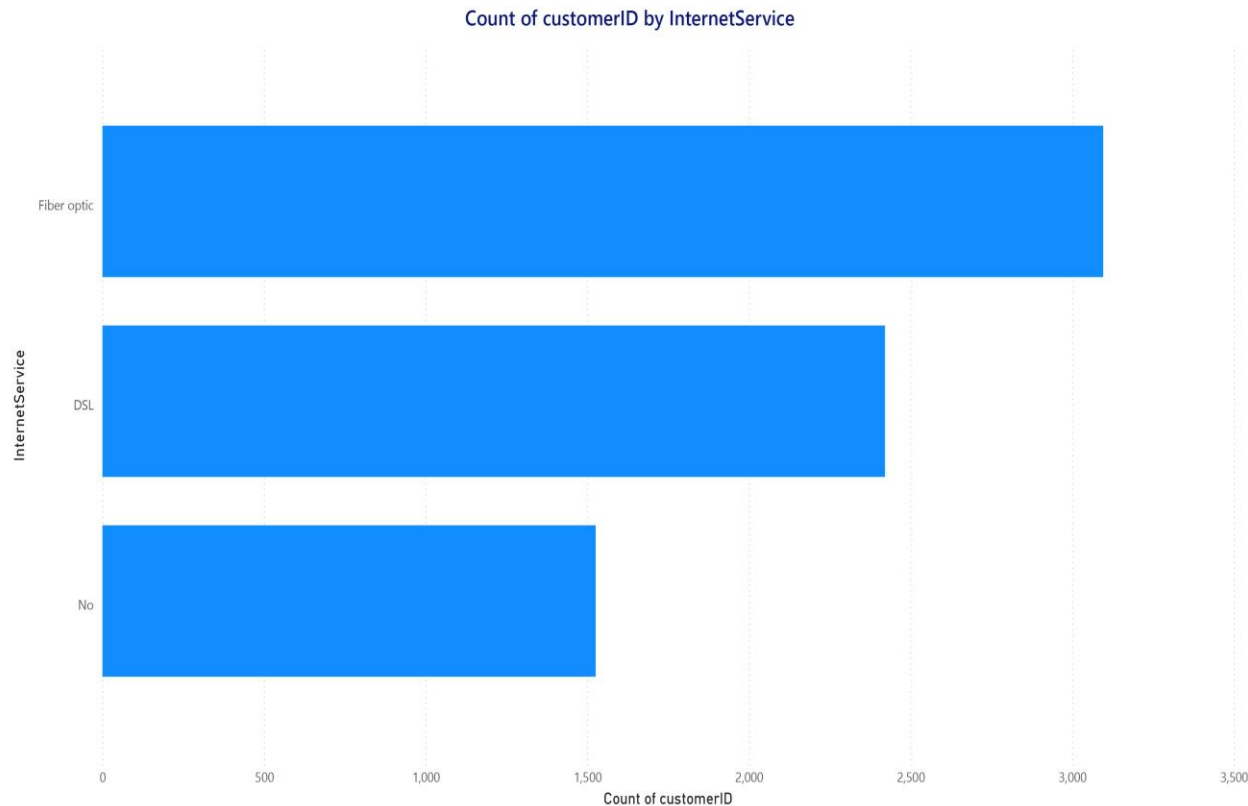


- 55% of customers are on month-to-month contracts.
- Approximately 44% use fiber optic internet service.
- The customer base includes a mix of short- and long-tenure customers.
- Monthly charges are concentrated in a lower-to-middle range, while TotalCharges is strongly linked to tenure.

Highest-Churn Segments

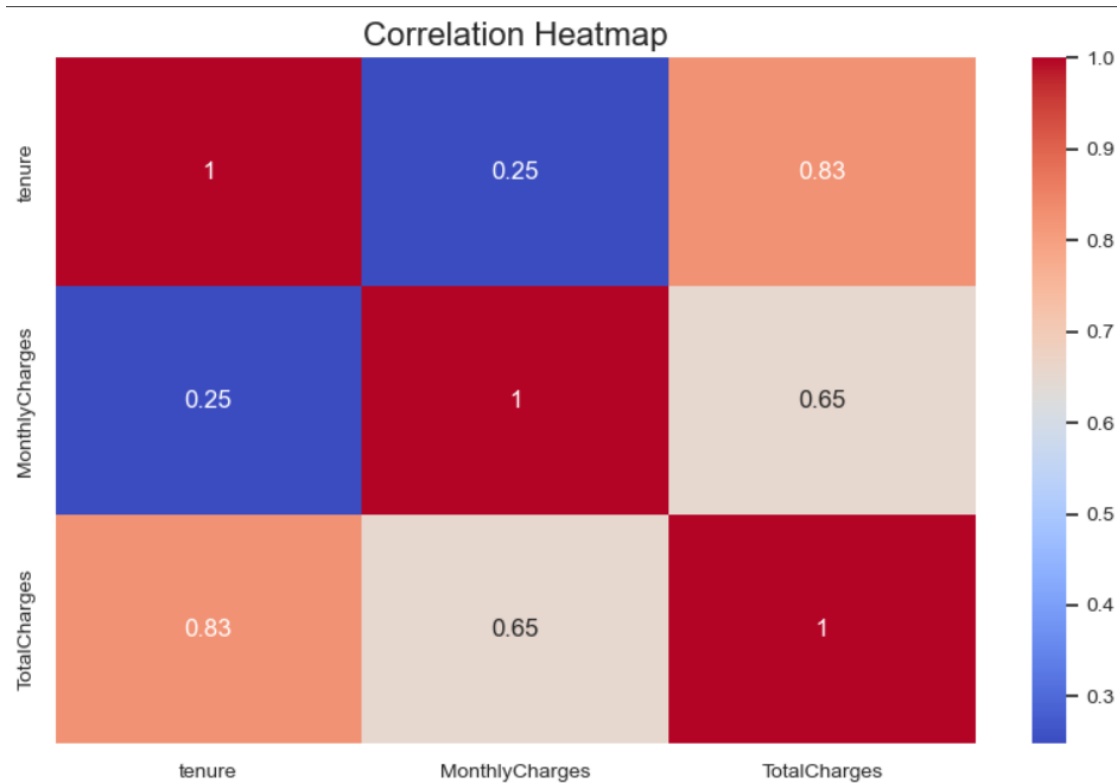
- Month-to-month contract customers have the highest churn rate.
- Customers in the 0–12 month tenure group have the highest churn rate.
- Fiber optic customers show significantly higher churn than other internet-service categories.
- Electronic check customers show substantially higher churn.
- These segments should be prioritized for deeper root-cause investigation and targeted retention.

Services Associated with Churn



- Phone service shows only a small difference in churn between customers with and without the service.
- Multiple-line customers show slightly higher churn.
- Customers without Online Security, Online Backup, Device Protection and Tech Support show much higher churn.
- Customers without Streaming TV or Streaming Movies show slightly higher churn.
- The service findings show association, not proof that a service causes or prevents churn.

Contract, Tenure and Charges



- Contract type is strongly associated with retention: month-to-month customers are the most exposed to churn.
- Short tenure is associated with higher churn, making the first 12 months a critical retention window.
- Correlation: tenure vs TotalCharges ≈ 0.83 ; MonthlyCharges vs TotalCharges ≈ 0.65 ; tenure vs MonthlyCharges ≈ 0.25 .

Business Risks and Opportunities

- RISKS: continued early-life churn, heavy exposure among month-to-month customers; elevated churn in fiber/electronic-check segments.
- OPPORTUNITIES: strengthen onboarding; convert suitable customers to longer contracts; improve service/support adoption.
- OPPORTUNITIES: use payment behaviour and service configuration to prioritize retention outreach.
- RISK CONTROL: investigate the underlying reasons before assuming that any service category is itself the cause of churn.

RECOMMENDATIONS FOR MANAGEMENT

1. Launch an early-life retention programme focused on customers in their first 12 months.
2. Create targeted incentives for suitable month-to-month customers to move to longer contracts.
3. Investigate fiber-optic churn through pricing, network quality, complaints and onboarding data.
4. Promote relevant support/security/protection bundles and measure their effect on retention.
5. Review the electronic-check payment journey and encourage reliable alternative payment methods where appropriate.
6. Establish a recurring churn dashboard and monitor churn by tenure, contract, service and payment method.